Comprehensive Hypertrophy Training Manual & Exercise Handbook

An evidence-based guide to programming execution, warm-up protocols, movement substitutions, and hypertrophy fundamentals.

Resources & Tooling

- **Nutrition & Macronutrient Calculator**: Access your personalized calorie and macronutrient breakdown targets via the nutrition calculator portal.
- **App Walkthrough Guide**: Review the training application tutorial video to familiarize yourself with logging workouts, tracking sets, and navigating your routine.

Starting Mid-Block: Onboarding & Acclimation Protocol

If you are beginning your program after a training block is already underway:

- **Current Training Match**: If your prior routine closely mirrors the current block's volume, intensity, and movement selection, you may jump directly into the scheduled week.
- **New Training Style / Acclimation**: If the exercises or programming structure are unfamiliar, use your initial week as an introductory acclimation/deload phase.
- **Failure Management**: Regardless of which path you choose, refrain from taking sets to absolute failure during week one. This prevents premature systemic and muscular fatigue.
- **Acclimation Details**: When treating week one as an acclimation phase, avoid pushing excessive intensity. Focus on mastering movement mechanics and tempo. Aim for **3 to 4 Reps in Reserve (RIR)** on all working sets (leaving 3–4 reps before failure). If you are not familiar with RIR, simply ensure every set finishes well before mechanical failure.

Thanks to the repeated bout effect, this submaximal entry sharply curtails excessive delayed-onset muscle soreness and fatigue as your body adapts.

Fundamental Terminology

- **Eccentric (Negatives)**: The phase of a lift where the target muscle lengthens under tension (e.g., descending into a squat, lowering the bar toward your chest on a bench press, or the upward return on a lat pulldown).
- **Isometric**: Holding a static position while actively maintaining force production, typically executed either at full peak contraction or in the fully lengthened stretch.
- **AMRAP**: "As Many Reps As Possible" completed with strict technical execution.
- **Drop Set**: Completing a prescribed set to the target rep threshold, immediately dropping the resistance by 20–30%, and continuing the exercise to failure without resting.
- **Progressive Overload**: Systematically increasing the mechanical demands on your muscular system over time, primarily achieved by adding load, increasing repetitions, or improving execution quality.
- **Deload**: A planned reduction in training stress designed to dissipate accumulated systemic and localized fatigue. Deloads can be achieved by decreasing resistance, dialing back total set volume, leaving more reps in reserve, or altering training stimuli.
- **Tempo**: Expressed as a 4-digit sequence in seconds, read left to right:
    a. *First number*: Duration of the eccentric (lowering/lengthening) phase.
    b. *Second number*: Pause in the stretched/lengthened position (0 indicates no pause).
    c. *Third number*: Speed of the concentric (lifting/shortening) phase (typically explosive/1 second).
    d. *Fourth number*: Pause at full muscular contraction/shortening.
- **Superset**: Performing two exercises back-to-back with minimal to no rest between them.

Warm-Up Protocol & Ramp-Up Sets

The most specific and effective warm-up is performing lighter preparation sets using the actual exercise scheduled:

1. **Optional General Warm-Up**: If you have been inactive or seated prior to your session and feel physically stiff, perform 10 minutes of low-intensity walking (e.g., 2.0–3.5 mph on a treadmill) before touching any weights.
2. **Ramp-Up Structure**: Complete 1 to 3 non-taxing warm-up sets prior to your working sets. Do not push any warm-up sets to failure.
    - **Warm-Up Set 1**: ~50% of your initial working load for 6–15 repetitions. Focus on a deep stretch, deliberate control, and a solid peak contraction. If you feel tight, slow down the eccentric and add a 1-second pause at both the stretched and contracted endpoints.
    - **Warm-Up Set 2**: 65–70% of working weight. Match the rep cadence and tempo of your working sets. Rest briefly—only until your heart rate and respiration normalize.
    - **Warm-Up Set 3**: 80–90% of working weight for 1–5 crisp reps.
    - *Adjustments*: If transitioning between large jumps in load, you may insert an intermediate set bridging 10–15% increments. Adjust total warm-up sets based on the lift and how warmed up you feel.

Exercise Substitutions Directory

If your gym lacks specific equipment, substitute with a movement that shares a nearly identical resistance profile and joint mechanic:

Glutes

Prescribed Exercise	Approved Substitutions
45-Degree Back Extensions	Barbell glute bridges, hip thrusts, cable glute-max kickbacks
Donkey Kickback	Cable kickbacks, Smith machine kickbacks, barbell bridges/thrusts, 45-degree back extensions
Glute Bridge	45-degree back extensions, hip thrust variations, glute-ham raises (GHR)
Unilateral Leg Press (Glute-Biased)	Glute-biased split squats or lunges, glute-biased barbell/Smith machine squats

Prescribed Exercise	Approved Substitutions
Glute Medius Kickback	Resistance band kickbacks, hip circle kickbacks, seated hip abductor machine (leaning back)

Quadriceps

Prescribed Exercise	Approved Substitutions
Hack Squat / Pendulum Squat	Heel-elevated SSB/barbell squats, barbell back squats, quad-biased leg press, quad-biased split squats, heel-elevated Smith machine squats
Leg Extensions	Cable leg extensions, resistance band leg extensions
Bilateral Leg Press (Quad-Biased)	Quad-biased split squats, forward lunges (in-place), Smith machine split squats or back squats
Sissy Squat	Reverse Nordics, lean-back leg extensions (minimizing hip flexion)

Hamstrings

Prescribed Exercise	Approved Substitutions
Seated Hamstring Curls	Lying leg curls, banded dumbbell lying leg curls, any knee-flexion exercise
Stiff-Leg / Hamstring-Biased RDL	Good mornings (Dumbbell, Barbell, or Trap Bar)

Adductors

Prescribed Exercise	Approved Substitutions
Adductor Machine	Copenhagen planks, cable hip adductions, banded hip adductions

Chest (Pectorals)

Prescribed Exercise	Approved Substitutions
Bilateral Clavicular Cable Press	Single-arm press-around, incline chest press machine, Smith machine incline press, dumbbell incline press, barbell incline bench

Prescribed Exercise	Approved Substitutions
Bilateral Sternal Cable Press	Pec deck fly, flat chest press machine, sternal cable fly, Smith machine flat press
Flat Dumbbell Bench Press	Flat barbell bench press, flat chest press machine, converging flat press, Smith machine flat bench
Incline Dumbbell Bench Press	Incline cable press, machine incline press, converging incline machine press, Smith machine incline press
Pec Deck / Chest Fly	Mid-to-low chest cable fly/press, flat chest press machine, dumbbell flys, flat dumbbell press
Upper Chest Cable Press	Dumbbell incline bench, Smith machine incline bench, incline machine chest press

Shoulders (Deltoids)

Prescribed Exercise	Approved Substitutions
Dumbbell Anterior Delt Press	Overhead converging shoulder press machine, standard overhead machine press, Smith machine shoulder press, standing/seated cable overhead press
Chest-Supported DB Lateral Raise	Cable lateral raises, seated dumbbell lateral raises, lateral raise machine
Chest-Supported DB Y-Raise	Cable Y-raises
Hand-Height Cable Lateral Raise	Incline bench lying dumbbell lateral raises, leaning dumbbell lateral raises
Rear Delt Fly	Cable rear delt pull-arounds (slight arm bend), chest-supported dumbbell rear delt flys
Dumbbell Front Raise	Cable front raises

Back

Prescribed Exercise	Approved Substitutions
Cable Pullover	Straight-arm lat pulldowns, standard lat pulldowns
Seated Lat-Biased Row	Step-back hammer/cable row, kneeling single-arm cable row (behind line of pull), chest-

Prescribed Exercise	Approved Substitutions
	supported dumbbell row (elbows close to torso)
Lat-Biased Pulldown	Bilateral neutral-grip pulldowns (D-handles preferred over straight bars), kneeling high-to-low Hammer Strength row/pulldown
T-Bar Row	Seated cable row, chest-supported barbell or dumbbell row, chest-supported machine rows with equivalent arm path
Frontal Plane Pulldown	Wide-grip straight-bar pulldown, assisted pull-ups, dual-cable pulldowns, diverging machine pulldown
Upper Back Row (45° & 90° path)	Any chest-supported row matching prescribed arm abduction angle (dumbbells, barbells, seated cable rows, or machines)
Kelso Shrugs	Chest-supported row machines/benches, dumbbell shrugs, Smith machine shrugs, barbell shrugs, standing machine shrugs

Arms & Forearms

Prescribed Exercise	Approved Substitutions
Seated Overhead Triceps Extension	JM press, overhead cable triceps extension
Dips	Overhead cable extensions, JM press, dip machine, bodyweight skull crushers, lying barbell/dumbbell skull crushers
Long-Rope / Cross-Cable Extension	Resistance band pushdowns, dual-cable pushdowns
Face-Away Cable Curl	Seated strict dumbbell curl (incline or flat), EZ-bar curl
Seated Supinated Dumbbell Curl	Cable curls, machine preacher/bicep curls
Zottman Curl	Alternating supinated curls, reverse grip curls
EZ-Bar Reverse Curl	Cable reverse curls, dumbbell reverse curls
Hammer Curl	Cable rope hammer curls, neutral-grip dumbbell curls
Standing Wrist Flexion Curl	Bench-supported wrist curls (flexion/extension), cable wrist curls, behind-the-back barbell wrist

Prescribed Exercise	Approved Substitutions
	curls

Abdominals

Prescribed Exercise	Approved Substitutions
Cable Crunch	Floor crunches using an abdominal mat, decline bench crunches, Garhammer raises

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Frequently Asked Questions (FAQ)

How do I accurately gauge progress?

1. **Progressive Overload**: Muscle building takes time, but your performance metrics provide direct feedback. Tracking your training log and seeing steady improvements in load, rep counts, and mechanical execution week over week is the most reliable indicator of adaptation.
2. **Body Composition Assessments**: Scale weight alone is incomplete. Regular physique check-in photos and consistent waist circumference measurements provide clarity. If your objective is fat loss and scale weight stays flat or drifts up, yet your waist measurement decreases and your muscle definition improves, you are gaining lean mass while shedding fat.

How much muscle can I realistically expect to build?

Your rate of muscular hypertrophy depends on your training age, genetics, and how rigorously you optimize your training stimulus, nutrition, and recovery. Rather than fixating on a specific numerical target, maximize execution across those core pillars. Doing so guarantees you achieve the best rate of growth possible for your physiology.

Should I work out when I am still sore?

If residual muscle soreness noticeably impairs your ability to execute lifts with proper technique or output force, you will benefit more from inserting an extra rest day than forcing the workout. Do not panic about falling "off schedule"—individual recovery timelines vary. Training through

deep muscular trauma creates unproductive fatigue and blunts adaptation. Building muscle is a long-term marathon; take recovery when needed.

How should I determine my working weights?

Use your first one to two weeks to calibrate appropriate loads for each prescribed repetition bracket. If you are already accustomed to a movement, rely on your past logs. If an exercise is unfamiliar, ramp up progressively across warm-ups until you find a resistance that places you at the designated proximity to failure within the target rep range.

When and how do I increase the weight on working sets?

When you successfully hit the upper ceiling of a prescribed rep range with clean form, advance your load:

- If you hit the top of the rep target on an early set with relative ease, increase the load by 5–10% for the subsequent sets.
- If hitting the top rep count was an intense grind, keep the load identical for the remaining sets and focus on hitting that target again next workout, or strive to add a single rep across your sets before adding weight.

Can I shuffle my workout days around?

The training split is carefully arranged so that training sessions and recovery periods synergize. Shuffling workout days randomly risks overlapping muscular stress and under-recovering prime movers. If your schedule requires an adjustment, add an extra rest day and continue the workouts in their designated order rather than restructuring the sequence.

What if I rarely or never experience muscle soreness?

Delayed-onset muscle soreness is not a mandatory prerequisite for hypertrophy. However, if you never experience even mild soreness or fatigue, re-evaluate your proximity to failure and set intensity. If you are pushing your sets close to true mechanical limits and advancing performance, you should periodically experience subtle soreness.

Is dedicated mobility work necessary?

Strength training through a full active range of motion under load is one of the most effective mobility modalities available. Lifting properly conditions your joints, tendons, and muscles

through deep lengthened positions. If you wish to incorporate supplementary static stretching, schedule it post-workout or on active recovery days.

Is this program tailored for fat loss, muscle growth, or body recomposition?

This program functions effectively for cutting, lean gaining, or recomposition. Hypertrophy-focused resistance training is identical whether you are eating in a caloric deficit, surplus, or maintenance. Your dietary intake dictates whether you lose adipose tissue or build muscle, while the training ensures muscle mass is preserved or built.

When should I wear a lifting belt?

A weightlifting belt should be utilized on heavy compound exercises that require intense intra-abdominal pressure and spinal bracing (such as heavy squats or deadlift variations). If you have not mastered the mechanics of intra-abdominal bracing, learn the technique first before relying on a belt. A belt is unnecessary on light isolations (e.g., standing arm curls).

When should I use lifting straps?

Use lifting straps whenever grip endurance limits your performance before the target muscle gives out. Pulling exercises—such as heavy rows, Romanian deadlifts (RDLs), and lat pulldowns—frequently tax grip before the lats or hamstrings reach failure. Straps eliminate grip as a bottleneck, ensuring full stimulus to the intended muscle group.